A Rigorous Galerkin Closure for the Elastodynamic Volume Integral Equation on a Cube

27-Component Path-B Formulation

Research Notes

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1 Motivation: differentiating under the integral fails

The classical moment-method closure for the cube T-matrix begins from the Lippmann–Schwinger (LS) equation

$$u_i(\boldsymbol{r}) \;=\; u_i^0(\boldsymbol{r}) + \omega^2 \int_V G_{ij}(\boldsymbol{r}-\boldsymbol{r}')\,\delta\rho(\boldsymbol{r}')\,u_j(\boldsymbol{r}')\,dV' - \int_V \partial'_k G_{in}(\boldsymbol{r}-\boldsymbol{r}')\,\delta c_{nklj}(\boldsymbol{r}')\,\partial'_l u_j(\boldsymbol{r}')\,dV', \quad (1)$$

where G_{ij} is the elastostatic Green's tensor and $V \subset \mathbb{R}^3$ is the inclusion. To obtain a closed finite-dimensional system one then formally takes derivatives at the centre $\boldsymbol{r} = \boldsymbol{0}$:

$$\begin{aligned}
u_i(\boldsymbol{0}) &= u_i^0(\boldsymbol{0}) + \omega^2 \int_V G_{ij}\,\delta\rho\,u_j\,dV' - \int_V \partial'_k G_{in}\,\delta c_{nklj}\,\partial'_l u_j\,dV', \\
\partial_p u_i(\boldsymbol{0}) &= \partial_p u_i^0(\boldsymbol{0}) - \omega^2 \int_V \partial'_p G_{ij}\,\delta\rho\,u_j\,dV' + \int_V \partial'_p\partial'_k G_{in}\,\delta c_{nklj}\,\partial'_l u_j\,dV', \qquad (2) \\
\partial_q\partial_p u_i(\boldsymbol{0}) &= \partial_q\partial_p u_i^0(\boldsymbol{0}) + \omega^2 \int_V \partial'_q\partial'_p G_{ij}\,\delta\rho\,u_j\,dV' - \int_V \partial'_q\partial'_p\partial'_k G_{in}\,\delta c_{nklj}\,\partial'_l u_j\,dV',
\end{aligned}$$

where the r–derivatives have been converted to r'–derivatives by translation invariance $\partial_r G(\boldsymbol{r} - \boldsymbol{r}') = -\partial_{r'} G(\boldsymbol{r} - \boldsymbol{r}')$.

The Leibniz rule used in passing from (1) to (2) requires (DCT version) a dominating function $|\partial_r f(\boldsymbol{r}, \boldsymbol{r}')| \leq g(\boldsymbol{r}')$ with $g \in L^1(V)$. The static elastic kernel has the singularity

$$G_{ij}^{\text{stat}}(\boldsymbol{s}) \sim \frac{1}{|\boldsymbol{s}|}, \quad \partial' G \sim \frac{1}{|\boldsymbol{s}|^2}, \quad \partial'\partial' G \sim \frac{1}{|\boldsymbol{s}|^3}, \quad \partial'\partial'\partial' G \sim \frac{1}{|\boldsymbol{s}|^4}, \tag{3}$$

so already at the level of $\partial' G$ *no integrable dominating function exists* when $\boldsymbol{r}$ is in the support of the inclusion. Each of the equations in (2) therefore makes only *formal* sense; converting them into rigorous statements requires distributional regularization (Cauchy principal value, Hadamard finite part, and Eshelby-type delta corrections).

LS derivative order	worst kernel	status
0 (body force)	$1/\|\boldsymbol{s}\|$	L^1_{loc} ✓
0 (elastic kernel)	$1/\|\boldsymbol{s}\|^2$	conditionally convergent (PV)
1 (body force)	$1/\|\boldsymbol{s}\|^2$	PV
1 (elastic kernel)	$1/\|\boldsymbol{s}\|^3$	**Eshelby delta required**
2 (body force)	$1/\|\boldsymbol{s}\|^3$	**Eshelby delta required**
2 (elastic kernel)	$1/\|\boldsymbol{s}\|^4$	**derivative-of-delta / Hadamard PF**

Table 1: Singular structure of the formal closure equations (2).

2 Path B: Galerkin formulation on a polynomial trial space

We replace point evaluation by a Galerkin (weak) projection onto a finite-dimensional polynomial trial space. The resulting bilinear forms involve only G_{ij} itself (the singularity $1/|\boldsymbol{s}|$ is locally L^1 in $\mathbb{R}^3$), so *every entry of the closure matrix is an absolutely convergent integral.* No principal values, no delta corrections, no distributional bookkeeping.

2.1 Trial space T_{27}

Let $V = [-a, a]^3$. Denote by $P_2(V)$ the space of real polynomials in $\boldsymbol{r} = (r_1, r_2, r_3)$ of total degree ≤ 2, $\dim P_2(V) = 10$. The naive vector trial space $(P_2(V))^3$ has dimension 30. We remove the 3 rigid-rotation modes by symmetrizing the linear part, giving the 27-dimensional subspace

$$T_{27} \;=\; \left\{ \boldsymbol{\varphi}(\boldsymbol{r}) \;\middle|\; \varphi_i(\boldsymbol{r}) = U_i + \varepsilon_{ai}\, r_a + \tfrac{1}{2}\, M_{abi}\, r_a r_b, \quad \varepsilon_{ai} = \varepsilon_{ia}, \quad M_{abi} = M_{bai} \right\}. \tag{4}$$

The dimensions add as $3 + 6 + 18 = 27$, matching one displacement monopole, the symmetric strain tensor, and the symmetric second-derivative tensor of $\boldsymbol{u}$ (equivalent to the strain gradient $\partial_k \varepsilon_{ij}$ through the kinematic identity $\partial_k \partial_l u_m = \partial_k \varepsilon_{lm} + \partial_l \varepsilon_{km} - \partial_m \varepsilon_{kl}$).

2.2 Galerkin LS

Definition 1 (Path-B closure)**.** The Path-B 27-component closure is the variational problem: find $\boldsymbol{u}^h \in T_{27}$ such that for all $\boldsymbol{\varphi} \in T_{27}$,

$$\langle \boldsymbol{\varphi}, \boldsymbol{u}^h \rangle_V \;=\; \langle \boldsymbol{\varphi}, \boldsymbol{u}^0 \rangle_V + \omega^2 \delta\rho\, \mathcal{B}_{\text{body}}[\boldsymbol{\varphi}, \boldsymbol{u}^h] - \delta c_{nklj}\, \mathcal{B}_{\text{el}}[\boldsymbol{\varphi}, \boldsymbol{u}^h], \tag{5}$$

where the bilinear forms are

$$\langle \boldsymbol{\varphi}, \boldsymbol{u} \rangle_V := \int_V \varphi_i(\boldsymbol{r})\, u_i(\boldsymbol{r})\, dV, \tag{6}$$

$$\mathcal{B}_{\text{body}}[\boldsymbol{\varphi}, \boldsymbol{u}] := \int_V\!\!\int_V \varphi_i(\boldsymbol{r})\, G_{in}(\boldsymbol{r}-\boldsymbol{r}')\, u_n(\boldsymbol{r}')\, dV\, dV', \tag{7}$$

$$\mathcal{B}_{\text{el}}[\boldsymbol{\varphi}, \boldsymbol{u}] := \int_V\!\!\int_V \varphi_i(\boldsymbol{r})\, \partial'_k G_{in}(\boldsymbol{r}-\boldsymbol{r}')\, \partial'_l u_j(\boldsymbol{r}')\, dV\, dV'. \tag{8}$$

2.3 Why this is well defined

Proposition 2. *For polynomial $\boldsymbol{\varphi}, \boldsymbol{u} \in T_{27}$, every bilinear form in (5) is an absolutely convergent integral. No distributional regularization is needed.*

Proof sketch. The mass form is trivial because the integrand is a polynomial. For $\mathcal{B}_{\text{el}}$, we use integration by parts on $\boldsymbol{u}$ (legitimate because $\boldsymbol{u}$ is a polynomial and the boundary term is a 2D integral against a smooth L^1_{loc} kernel):

$$\mathcal{B}_{\text{el}}[\boldsymbol{\varphi}, \boldsymbol{u}] = -\int_V\!\!\int_V \varphi_i\, G_{in}\, \partial'_k \partial'_l u_j\, dV'\, dV + \int_V \!\!\oint_{\partial V} \varphi_i\, G_{in}\, \partial'_l u_j\, n_k\, dS'\, dV. \tag{9}$$

Both integrands now contain only $G_{in}(\boldsymbol{r}-\boldsymbol{r}') \sim 1/|\boldsymbol{r}-\boldsymbol{r}'|$ multiplied by polynomials/smooth functions. The 6-fold integral $\iint_{V\times V} 1/|\boldsymbol{r}-\boldsymbol{r}'|\, dV\, dV'$ converges by the Hardy–Littlewood–Sobolev inequality (or directly: in $\mathbb{R}^6$ the singular set $\{\boldsymbol{r}=\boldsymbol{r}'\}$ has codimension 3 and $1/|\boldsymbol{r}-\boldsymbol{r}'|$ has order-1 singularity, hence is locally L^1 in $\mathbb{R}^6$). The 5-fold boundary integral is similar. The body form is even simpler since no integration by parts is needed. □ □

2.4 Equivalence to point-collocation in the limit

The classical Path-A closure (point evaluation at $\boldsymbol{r}=\boldsymbol{0}$) is the formal limit of a Petrov–Galerkin scheme with Dirac-delta test functions. Both Path A and Path B converge to the exact volume integral equation as the polynomial degree $\to \infty$; at finite truncation their stiffness matrices differ but encode the same physical operator. Path B is preferable for analysis because it is symmetric, energy conserving, and free of singular regularizations.

3 Center-of-mass / relative coordinates

The 6-fold integrals in (5) are reduced to 3-fold integrals against the *cube autocorrelation* via the *centre-of-mass and relative coordinate* change of variables (also called *sum-and-difference variables* in the Eshelby/Mura literature, or *Jacobi coordinates* in the simplest 2-body case):

$$\boldsymbol{\xi} = \tfrac{1}{2}(\boldsymbol{r}+\boldsymbol{r}'), \qquad \boldsymbol{s} = \boldsymbol{r}-\boldsymbol{r}', \qquad \boldsymbol{r} = \boldsymbol{\xi} + \tfrac{\boldsymbol{s}}{2}, \quad \boldsymbol{r}' = \boldsymbol{\xi} - \tfrac{\boldsymbol{s}}{2}, \qquad d\boldsymbol{r}\, d\boldsymbol{r}' = d\boldsymbol{\xi}\, d\boldsymbol{s}. \tag{10}$$

The constraint $\boldsymbol{r}, \boldsymbol{r}' \in V$ becomes $\boldsymbol{\xi} \in V \cap (V+\boldsymbol{s}/2) \cap (V-\boldsymbol{s}/2)$, which for the cube $V=[-a,a]^3$ simplifies to a centred box of half-lengths $b_i(\boldsymbol{s}) = a - |s_i|/2$:

$$\boldsymbol{\xi} \in B(\boldsymbol{s}) := \prod_{i=1}^{3} \big[-(a-|s_i|/2),\, a-|s_i|/2\big], \qquad \text{valid when } |s_i| < 2a. \tag{11}$$

The 6-fold integral becomes

$$\int_V\!\!\int_V P(\boldsymbol{r})\, K(\boldsymbol{r}-\boldsymbol{r}')\, Q(\boldsymbol{r}')\, d\boldsymbol{r}\, d\boldsymbol{r}' \;=\; \int_{[-2a,2a]^3} K(\boldsymbol{s}) \underbrace{\Big[\int_{B(\boldsymbol{s})} P\big(\boldsymbol{\xi}+\tfrac{\boldsymbol{s}}{2}\big)\, Q\big(\boldsymbol{\xi}-\tfrac{\boldsymbol{s}}{2}\big)\, d\boldsymbol{\xi}\Big]}_{\text{polynomial in } (\boldsymbol{s},a)} d\boldsymbol{s}. \tag{12}$$

For polynomial P, Q the inner integral is elementary: every ξ_i^p integrates over $[-b_i, b_i]$ to $\frac{2b_i^{p+1}}{p+1}$ when p is even and to 0 when p is odd, giving a polynomial in $(b_1, b_2, b_3) = (a - |s_1|/2,\ a - |s_2|/2,\ a - |s_3|/2)$.

Cube autocorrelation. For $P = Q = 1$ the inner integral is exactly the autocorrelation of the cube indicator,

$$(\chi_V \star \chi_V)(\boldsymbol{s}) \;=\; \prod_{i=1}^{3} \big(2a - |s_i|\big)_+, \tag{13}$$

the 3-D *tent function* $\Lambda_V(\boldsymbol{s})$.

4 Master integrals

After the centre-of-mass/relative reduction, every entry of the 27×27 Galerkin matrix is a finite linear combination of the following *master integrals* over the doubled cube $\Omega = [-2a, 2a]^3$:

$$\mathcal{I}_{p,q,r}(a) := \int_\Omega \frac{s_1^p\, s_2^q\, s_3^r}{|\boldsymbol{s}|}\, d\boldsymbol{s}, \tag{14}$$

$$\mathcal{J}_{p,q,r}^{(ij)}(a) := \int_\Omega \frac{s_i s_j\, s_1^p\, s_2^q\, s_3^r}{|\boldsymbol{s}|^3}\, d\boldsymbol{s}. \tag{15}$$

For the 9-component (current) closure only $p + q + r \le 3$ is needed; the 27-component closure requires $p + q + r \le 5$.

By the parity $\boldsymbol{s} \to -\boldsymbol{s}$ on each axis separately, $\mathcal{I}_{p,q,r}$ vanishes whenever any of p, q, r is odd, and $\mathcal{J}_{p,q,r}^{(ij)}$ vanishes unless the multiset $\{i, j, p\text{-copies of } 1, q\text{-copies of } 2, r\text{-copies of } 3\}$ has even count for each axis. This collapses the formally $\binom{8}{3} \cdot 3 = 168$ entries down to ~ 20 inequivalent integrals after cubic symmetry.

4.1 Tier-1 results (computed analytically)

The basic integral $\mathcal{I}_{0,0,0}(a)$ is the dimensionless cube constant g_0 scaled by a^2. Mathematica's `Integrate` returns it in the form

$$\int_{[-1,1]^3} \frac{1}{|\boldsymbol{u}|}\, du \;=\; -2\Big(\pi + \log\big(1351 - 780\sqrt{3}\big)\Big), \tag{16}$$

which equals the Pell-stable form used in the existing `CubeAnalytic.wl` package via the identity $1351 - 780\sqrt{3} = (70226 + 40545\sqrt{3})^{-2/3}$ (both sides are Pell units of $\mathbb{Z}[\sqrt{3}]$, satisfying $1351^2 - 3 \cdot 780^2 = 1$ and $70226^2 - 3 \cdot 40545^2 = 1$). Numerical agreement to $\sim$78 digits has been verified.

4.2 Verified analytical master integrals

The verification at degree 0 was reported in Section 4.1. We extend the table to include all parity-allowed master integrals up to degree 6:

The same family of integrals over the positive octant $[0, 1]^3$ supplies the body-bilinear-form building blocks; we collect them in Table 3.

4.3 Body bilinear-form building blocks

For the constant subspace of T_{27} the diagonal and off-diagonal entries of the body bilinear form $\mathcal{B}_{\text{body}}$ reduce, after the centre-of-mass / relative-coordinate substitution and an octant-symmetry

(p,q,r)	$\mathcal{M}_{p,q,r} = \int_{[-1,1]^3} \frac{x^p y^q z^r}{\lvert \boldsymbol{r} \rvert}\, dV$
$(0,0,0)$	$-2\pi - \frac{4}{3}\log\left(70226 - 40545\sqrt{3}\right)$
$(2,0,0)$	$\frac{1}{9}\left(6\sqrt{3} - \pi + \log\left(3650401 + 2107560\sqrt{3}\right)\right)$
$(2,2,0)$	$\frac{1}{135}\left(108\sqrt{3} + 18\pi + 63\log(2) + 20\log\left(26 - 15\sqrt{3}\right) - 60\log\left(\sqrt{3}-1\right) - 66\log\left(1+\sqrt{3}\right) - 90\coth^{-1}\right.$
$(4,0,0)$	$-\frac{2}{45}\left(12\sqrt{3} + 7\pi - 42\log(2) + 99\log\left(\sqrt{3}-1\right) - 15\log\left(1+\sqrt{3}\right)\right)$
$(2,2,2)$	$\frac{1008\sqrt{3}-42\pi+522\log(2)+441\log\left(\sqrt{3}-1\right)-1395\log\left(1+\sqrt{3}\right)-84\log\left(2+\sqrt{3}\right)-90\log\left(11\sqrt{3}-19\right)+35\log\left(26+15\sqrt{3}\right)-672\coth^{-1}}{1260}$
$(4,2,0)$	$\frac{1}{630}\left(-48\sqrt{3} + 17\pi + 12\tanh^{-1}\left(\frac{21252634831\sqrt{3}}{36810643322}\right)\right)$
$(6,0,0)$	$\frac{1}{42}\left(-5\pi - 4\log(2) + 8\left(3\sqrt{3} + \log\left(5+3\sqrt{3}\right)\right)\right)$

Table 2: Closed-form values of the scalar master integrals $\mathcal{M}_{p,q,r}$ over the unit cube. Cubic symmetry equates entries that are permutations of each other; only canonical orderings $p \geq q \geq r$ are listed.

(p,q,r)	$\mathcal{M}^{+}_{p,q,r} = \int_{[0,1]^3} \frac{x^p y^q z^r}{\lvert \boldsymbol{r} \rvert}\, dV$
$(0,0,0)$	$-\frac{\pi}{4} - \frac{1}{6}\log\left(70226 - 40545\sqrt{3}\right)$
$(1,0,0)$	$\frac{1}{36}\left(-2\pi - 12\left(\sqrt{2} - \sqrt{3} + \sinh^{-1}(1)\right) + \cosh^{-1}(189750626) + 18\coth^{-1}\left(\sqrt{3}\right)\right)$
$(1,1,0)$	$\frac{1}{12}\left(1 - 7\sqrt{2} + 6\sqrt{3} + \log\left(5\sqrt{2} - 7\right) - 3\log\left(2-\sqrt{3}\right)\right)$
$(2,0,0)$	$\frac{1}{144}\left(12\sqrt{3} - 2\pi + \log\left(26650854921601 + 15386878263120\sqrt{3}\right)\right)$
$(1,1,1)$	$\frac{1}{5}\left(1 - 4\sqrt{2} + 3\sqrt{3}\right)$
$(2,1,0)$	$\frac{-504\sqrt{2}+672\sqrt{3}+32\pi+72\log\left(\sqrt{3}-1\right)-252\log\left(1+\sqrt{3}\right)+90\log\left(2\left(2+\sqrt{3}\right)\right)-261\sinh^{-1}(1)+40\cosh^{-1}(26)+45\coth^{-1}\left(\sqrt{2}\right)}{2880}$

Table 3: Closed-form values of the positive-octant master integrals $\mathcal{M}^{+}_{p,q,r}$ over $[0,1]^3$. Up to canonical ordering $p \geq q \geq r$ and $p+q+r \leq 3$ they exhaust all polynomial weights needed by the Path-B body bilinear form on the constant-and-linear subspace.

restriction, to three positive scalars. Define

$$i_A := \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\,\frac{1}{\lvert \boldsymbol{u} \rvert}\, d\boldsymbol{u}, \tag{17}$$

$$i_{B,11} := \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\,\frac{u_1^2}{\lvert \boldsymbol{u} \rvert^3}\, d\boldsymbol{u}, \tag{18}$$

$$i_{B,12} := \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\,\frac{u_1 u_2}{\lvert \boldsymbol{u} \rvert^3}\, d\boldsymbol{u}. \tag{19}$$

The first reduces by linearity to a signed combination of $\mathcal{M}^{+}_{p,q,r}$ values (Table 3):

$$i_A \;=\; \mathcal{M}^{+}_{0,0,0} \;-\; 3\mathcal{M}^{+}_{1,0,0} \;+\; 3\mathcal{M}^{+}_{1,1,0} \;-\; \mathcal{M}^{+}_{1,1,1} \;\approx\; 0.2352890805487075. \tag{20}$$

The remaining two are obtained by integrating once by parts in the u_1 direction; the singular kernel $\lvert \boldsymbol{u} \rvert^{-3}$ collapses to a $\lvert \boldsymbol{u} \rvert^{-1}$-class kernel and a clean two-dimensional residual:

$$i_{B,11} \;=\; i_A \;-\; J_{11}, \qquad J_{11} \;=\; \int_{[0,1]^2} (1-u_2)(1-u_3)\Big[\sqrt{1+u_2^2+u_3^2} - \sqrt{u_2^2+u_3^2}\Big]\, du_2\, du_3, \tag{21}$$

$$i_{B,12} \;=\; \int_{[0,1]^2} (1-u_2)(1-u_3)\, u_2 \Big[\frac{1}{\sqrt{u_2^2+u_3^2}} \;-\; \operatorname{arcsinh}\Big(\frac{1}{\sqrt{u_2^2+u_3^2}}\Big)\Big]\, du_2\, du_3. \tag{22}$$

Mathematica evaluates both two-dimensional integrals in closed form:

$$J_{11} = \frac{1}{240}\left(8+8\sqrt{2}-16\sqrt{3}-55\log(2)+15\log\left(3+2\sqrt{2}\right)+110\log\left(1+\sqrt{3}\right)-15\log\left(2+\sqrt{3}\right)-80\tan^{-} \right. \tag{23}$$

$$\approx 0.1568593870324717, \tag{24}$$

$$i_{B,12} = \frac{1}{240}\left(-8\sqrt{2}+4\sqrt{3}+20\log(2)+5\log(256)+40\log\left(1+\sqrt{2}\right)+22\log\left(3+2\sqrt{2}\right)-120\log\left(1+\sqrt{3}\right)\right) \tag{25}$$

$$\approx 0.04970171296728090. \tag{26}$$

A non-trivial consistency check is supplied by the trace identity

$$3\, i_{B,11} = \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\, \frac{u_1^2+u_2^2+u_3^2}{|\boldsymbol{u}|^3}\, d\boldsymbol{u} = i_A,$$

which is verified analytically and numerically to working precision.

4.4 The 9×9 body bilinear matrix on T_9

With the master integrals of Sections 4.2 and the three scalar building blocks i_A, $i_{B,11}$, $i_{B,12}$, the body bilinear form $\mathcal{B}_9 : T_9 \times T_9 \to \mathbb{R}$ is assembled entry-by-entry from

$$\mathcal{B}_9[i,j] = 32 \sum_{p,q=1}^{3} \Big\{ A\, \delta_{pq}\, K_1[\Pi_{ij,pq}] + B\, K_3[\Pi_{ij,pq};\, p,q] \Big\}, \tag{27}$$

where $\Pi_{ij,pq}(\boldsymbol{u})$ is the residual polynomial obtained by (a) substituting $\phi_p^{(i)}(\boldsymbol{r})\phi_q^{(j)}(\boldsymbol{r}')$, (b) integrating the centre-of-mass variable $\boldsymbol{\xi}$ over the overlap box, and (c) dividing out the standard tent factor $(1-u_1)(1-u_2)(1-u_3)$. The prefactor $32 = 4 \cdot 8/1$ absorbs the $s \mapsto 2u$ Jacobian 8, the Green-tensor rescaling $G(2\boldsymbol{u}) = \frac{1}{2}[A/|\boldsymbol{u}| + B\, u_p u_q/|\boldsymbol{u}|^3]$, and the symmetric-box factor 2/1. A further factor of 8 appears on parity-even entries when we restrict to the positive octant $[0,1]^3$, so a diagonal displacement-block entry is proportional to 256.

Under cubic symmetry the T_9-basis decomposes into four irreducible blocks, each of which is characterised by at most two independent scalars:

- **Displacement block** ($i,j \in \{1,2,3\}$): diagonal, with a single scalar.
- **Displacement–strain block** ($i \in \{1,2,3\}$, $j \in \{4,\dots,9\}$): identically zero by parity.
- **Axial-strain block** ($i,j \in \{4,5,6\}$): diagonal entry α, off-diagonal entry β.
- **Axial–shear block** ($i \in \{4,5,6\}$, $j \in \{7,8,9\}$): identically zero by parity.
- **Shear-strain block** ($i,j \in \{7,8,9\}$): diagonal entry γ, off-diagonals identically zero by parity.

Displacement block

All three displacement-block diagonal entries are equal and reduce in closed form to

$$\mathcal{B}_9[1,1] = \mathcal{B}_9[2,2] = \mathcal{B}_9[3,3] = 256\left(A\, i_A + B\, i_{B,11}\right) = 60.23400462046913\, A + 20.07800154015638\, B, \tag{28}$$

with off-diagonal entries vanishing by parity.

Axial-strain block

The diagonal entry $\alpha := \mathcal{B}_9[4,4] = \mathcal{B}_9[5,5] = \mathcal{B}_9[6,6]$ splits into

$$\alpha \;=\; \tfrac{256}{3}\Big[A\big(i_A - 2K_1^{(2,0,0)}\big) + B\big(i_{B,11} - 2K_3^{(2,0,0)}\big)\Big] \;=\; 15.37252824403423\,A + 3.949761457945144\,B, \tag{29}$$

where

$$K_1^{(2,0,0)} \;:=\; \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\,\frac{u_1^2}{|\boldsymbol{u}|}\,d\boldsymbol{u}, \quad K_3^{(2,0,0)} \;:=\; \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\,\frac{u_1^4}{|\boldsymbol{u}|^3}\,d\boldsymbol{u}.$$

The off-diagonal entry $\beta := \mathcal{B}_9[4,5]$ is purely a B-channel correction,

$$\beta \;=\; -256\,B\,K_{3,\mathrm{off}}^{(1,1,0)} \;=\; 0\,A \;-\; 1.471925680511371\,B, \tag{30}$$

where $K_{3,\mathrm{off}}^{(1,1,0)} = \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\,u_1 u_2\,\frac{u_1 u_2}{|\boldsymbol{u}|^3}\,d\boldsymbol{u}$. Note that β vanishes in the A-channel because the integrand of the A piece is odd in (u_1, u_2).

Shear-strain block

The diagonal entry $\gamma := \mathcal{B}_9[7,7] = \mathcal{B}_9[8,8] = \mathcal{B}_9[9,9]$ evaluates to

$$\gamma \;=\; \tfrac{128}{3}\Big[A\big(i_A - 2K_1^{(2,0,0)}\big) + B\big(i_{B,11} - 2K_3^{(0,2,0)}\big)\Big] - 128\,B\,K_{3,\mathrm{off}}^{(1,1,0)} \;=\; 7.686264122017116\,A + 2.11972885626658 \tag{31}$$

where $K_3^{(0,2,0)} = \int_{[0,1]^3} (1-u_1)(1-u_2)(1-u_3)\,\frac{u_2^2 u_1^2}{|\boldsymbol{u}|^3}\,d\boldsymbol{u}$. The relation $\gamma^{(A)} = \frac{1}{2}\,\alpha^{(A)}$ is an exact identity: the A-channel of γ is exactly half the A-channel of α, coming from the bilinearity of the two-component shear basis $\phi^{(7)} = (0, r_3/2, r_2/2)$.

Numerical verification of cubic symmetry (all six independent identities) was performed to 10^{-20} precision; see the Mathematica script output.

4.5 The 18×18 quad–quad block on T_{27}

Extending T_9 to T_{27} adjoins the 18 quadratic vector fields

$$\phi^{(9+6(k-1)+m)}(\boldsymbol{r}) \;=\; q_m(\boldsymbol{r})\,\boldsymbol{\varepsilon}_k, \qquad k \in \{1,2,3\},\;\; m \in \{1,\ldots,6\},$$

where the six quadratic monomials in canonical Voigt order are

$$q_1 = r_1^2, \quad q_2 = r_2^2, \quad q_3 = r_3^2, \quad q_4 = r_2 r_3, \quad q_5 = r_1 r_3, \quad q_6 = r_1 r_2.$$

We refer to $q_{1..3}$ as the squared (S) modes and to $q_{4..6}$ as the cross (X) modes. On the 18×18 quad–quad sub-block, the body bilinear form decomposes into an A-channel and a B-channel,

$$\mathcal{B}_{27}^{\mathrm{quad}}[i,j] \;=\; 32\,\big(a_{ij}\,A \;+\; b_{ij}\,B\big), \tag{32}$$

where the scalars a_{ij}, b_{ij} depend only on the orbit of (i,j) under the combined O_h action on the three coordinate axes and the $i \leftrightarrow j$ symmetry.

Enumerating all $18^2 = 324$ index pairs and partitioning by this action yields exactly **34 orbits**. Of these, **20 orbits** (total multiplicity 222) are identically zero in both channels: each such vanishing is a parity cancellation forced by cubic symmetry (the integrand is odd in at least one coordinate axis under the $[-1,1]^3$ integration). The remaining **14 orbits** (total multiplicity 102) carry nontrivial values, and are listed in Table 4.

Representative (i,j)	Structure	mult	A-value	B-value
$(10,10)$	S-S diag, axis-matched	3	17.126396135468	7.086377754856
$(11,11)$	S-S diag, axis-unmatched	6	17.126396135468	5.020009190302
$(10,11)$	S-S same-block off-diag, one matched	12	14.698679523967	5.444954908312
$(11,12)$	S-S same-block off-diag, both unmatched	6	14.698679523967	3.808769707337
$(13,13)$	X-X diag, no-factor-matched	3	3.988696969909	1.644075434839
$(14,14)$	X-X diag, one-factor-matched	6	3.988696969909	1.172310767534
$(11,17)$	S-X cross-block (type I)	12	0	0.692768591986
$(10,17)$	S-S cross-block (type Ia)	6	0	0.654189191338
$(11,16)$	S-S cross-block (type Ib)	6	0	0.654189191338
$(14,26)$	X-X cross-block (type I)	6	0	0.654189191338
$(13,20)$	X-X cross-block (type II)	6	0	−0.389578544262
$(14,19)$	X-X cross-block (type III)	6	0	−0.389578544262
$(10,21)$	S-X cross-block (type II)	12	0	−0.216414577671
$(11,21)$	S-X cross-block (type III)	12	0	−0.216414577671

Table 4: The 14 non-vanishing orbits of the 18×18 quad–quad block $\mathcal{B}_{27}^{\text{quad}}$. Indices i,j are global ($10 \le i,j \le 27$). The A- and B-values in each row are the two numerical coefficients of A and B in (32). Multiplicities sum to 102; the remaining 222 of the 324 entries sit in 20 zero orbits forced by cubic parity. Values are reported to ~ 12 significant digits from the `CubeT6Block.wl` `MachinePrecision`+ `PrecisionGoal`→ 12 assembly.

Independent scalars. The 14 non-zero orbits collapse to **3 distinct A-scalars and 10 distinct B-scalars**:

$$\alpha_1 \approx 17.126396135468, \qquad \alpha_2 \approx 14.698679523967, \qquad \alpha_3 \approx 3.988696969909,$$

$$\beta_1 \approx 7.086377754856, \qquad \beta_2 \approx 5.020009190302, \qquad \beta_3 \approx 5.444954908312,$$
$$\beta_4 \approx 3.808769707337, \qquad \beta_5 \approx 1.644075434839, \qquad \beta_6 \approx 1.172310767534,$$
$$\beta_7 \approx 0.692768591986, \qquad \beta_8 \approx 0.654189191338, \qquad \beta_9 \approx -0.389578544262,$$
$$\beta_{10} \approx -0.216414577671.$$

Across all 324 entries the observed within-orbit spread is 0 exactly in the A-channel and $\le 1.8 \times 10^{-15}$ (floating-point noise) in the B-channel, so these 13 scalars completely determine the 18×18 block. All values are exported to `Mathematica/CubeT6ScalarValues.wl`.

Symmetry structure and accidental identities. The A-channel vanishes on every cross-direction orbit ($k_i \neq k_j$) and on every S-X orbit: this reflects the fact that the A part of the elastic Green's tensor is proportional to $\delta_{ij}/|\mathbf{r}-\mathbf{r}'|$ (an isotropic trace piece), so it only connects modes carrying identical Cartesian vector components and with an even polynomial overlap. The B-channel, which carries the $(\mathbf{r}-\mathbf{r}')_i(\mathbf{r}-\mathbf{r}')_j/|\mathbf{r}-\mathbf{r}'|^3$ anisotropic piece, is nonzero on all 14 orbits.

An additional structural observation from Table 4 is that the 8 cross-direction orbits collapse to only *four* distinct B-values: $\beta_7 = 0.6928$, $\beta_8 = 0.6542$ (appearing in three orbits), $\beta_9 = -0.3896$ (appearing in two orbits), and $\beta_{10} = -0.2164$ (appearing in two orbits). These collapses are identities to ~ 12-digit precision that are not a direct consequence of the O_h-orbit decomposition alone: our canonical labelling distinguishes $(10,17)$, $(11,16)$, and $(14,26)$ as separate O_h orbits, yet they integrate to the same number. We conjecture that they reflect a deeper structural identity of the cube Green-tensor double integral; closed-form verification is left for future work.

Extraction method and precision. The scalars α_i, β_j were computed numerically by `Mathematica/CubeT6I` which assembles $\mathcal{B}_{27}^{\text{quad}}$ via `NIntegrate` at `MachinePrecision` with `PrecisionGoal`$\to 12$ on the corner-singular masters $\mathcal{M}^{+}_{p,q,r}$ and on the K_3 residual kernels. Higher `WorkingPrecision` values (20–25) gave per-row assembly times exceeding one hour; the current configuration completes all 324 entries in ~ 150 s and is cross-checked against the T_9 Tier-5 displacement diagonal (28) to better than 3×10^{-12}. The 7 exact high-degree masters $\mathcal{M}^{+}_{(2,2,2)}, \ldots, \mathcal{M}^{+}_{(8,0,0)}$ primed from `CubeT6Masters.wl` ensure that the leading parity-even contributions are closed-form. A dedicated overnight high-precision recomputation (`CubeT6OvernightHighPrec.wl`, 14 orbits at `WorkingPrecision`$\to 30$) is the next stage; it will supply the ≥ 20 digits needed for PSLQ-based closed-form reconstruction of the α_i and β_j in the same spirit as (29)–(31).

5 Validation plan: Path-A $\Leftrightarrow$ Path-B for the 9-component subsystem

The existing `CubeAnalytic.wl` package implements Path A with analytical Eshelby-delta corrections (the $(5\sqrt{3} - 2\pi)$ block from the old `GreensTensorScatteringPaper_Old.nb`). As a correctness test for the framework presented here, we will:

(V1) Compute the Path-B mass and stiffness matrices on the 9-dimensional subspace $T_9 \subset T_{27}$ spanned by constants and symmetric linear polynomials.

(V2) Solve the resulting 9×9 generalized eigenproblem for the effective contrasts $\Delta\rho^*, \Delta\lambda^*, \Delta\mu^*$.

(V3) Compare to the Path-A values produced by `compute_cube_tmatrix` in `cubic_scattering/effective_co`

Path A and Path B do not need to give identical matrix entries (they correspond to different finite-dimensional projections of the same operator), but the effective contrasts must agree to leading order in the polynomial truncation. Any discrepancy quantifies the truncation error and bounds the systematic error of Path A.

6 New integrals required for the 27-component closure

Beyond the constants used in the 9-component closure ($\mathcal{I}_{0,0,0}$, $\mathcal{J}^{(11)}_{0,0,0}$, $\mathcal{J}^{(11)}_{2,0,0}$, etc.), the 27-component closure introduces master integrals with total monomial degree up to 5. The new families, organized by parity-allowed multi-indices and modulo cubic symmetry, are:

(p,q,r) class	rank	needed for
$(2,2,0),\ (4,0,0)$	4	$\langle \boldsymbol{\varphi}^{(\text{lin})}, G\boldsymbol{\varphi}^{(\text{lin})} \rangle$ block
$(2,2,2),\ (4,2,0),\ (6,0,0)$	6	$\langle \boldsymbol{\varphi}^{(\text{quad})}, G\boldsymbol{\varphi}^{(\text{lin})} \rangle$ block
$(4,2,2),\ (4,4,0),\ (6,2,0),\ (8,0,0)$	8	$\langle \boldsymbol{\varphi}^{(\text{quad})}, G\boldsymbol{\varphi}^{(\text{quad})} \rangle$ block

Table 5: New cubic-symmetric master integral families needed for the 27-component Path-B closure. Each row corresponds to one block of the 27×27 stiffness matrix.

These integrals will be derived analytically in Section 4.1 and the subsequent stages of `CubeGalerkin27.wl`.

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